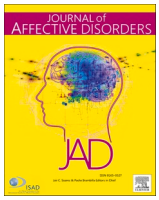
Journal of Affective Disorders 415 (2026) 122428



Contents lists available at ScienceDirect

Journal of Affective Disorders

journal homepage: www.elsevier.com/locate/jad



Research paper

Latent profiles of generative artificial intelligence use among Chinese college students: Associations with depression and anxiety

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ARTICLE INFO

Keywords:
Generative artificial intelligence
Latent profile analysis
Depression
Anxiety
College students

ABSTRACT

Background: While generative artificial intelligence (GenAI) has been rapidly adopted by college students, its relationship with mental health remains unclear. Most studies treat AI users as homogeneous, overlooking heterogeneity in GenAI use. This study identified latent profiles of GenAI use among Chinese college students and examined their associations with depression and anxiety.

Methods: A cross-sectional survey of 5748 Chinese college students assessed AI usage, motivations, AI literacy, dependency, and symptoms of depression and anxiety using the Beck Depression Inventory–II and Beck Anxiety Inventory. Latent profile analysis identified usage patterns. An exploratory random-forest classifier with SHapley Additive exPlanations (SHAP) evaluated whether these profiles could be predicted from psychological and demographic correlates not used for profile construction and identified key distinguishing factors.

Results: Four profiles were identified: Rational-Tool, Moderate-Recreational, Problem-Dependent, and Light-Exploratory. The Problem-Dependent profile (15%), characterized by high escapism motivation, low AI literacy, and high dependency, showed the highest depression and anxiety levels, whereas the Rational-Tool profile, characterized by high AI literacy and instrumental motivation, showed the most favorable outcomes. The profiles were recovered with good discrimination (accuracy = 0.81; macro-average AUC = 0.89), with the Problem-Dependent profile being the most distinguishable (AUC = 0.94). SHAP analysis identified depression, smartphone addiction, and executive-function difficulties as key correlates of the Problem-Dependent profile, whereas conscientiousness was central to the Rational-Tool profile.

Conclusions: GenAI use profiles were associated with different levels of depression and anxiety among college students. Person-centered prevention strategies should focus on usage motivation, AI literacy, and dependency rather than frequency alone.

1. Introduction

College years are characterized by the elevated risk of developing mental health issues. They coincide with emerging adulthood, the term introduced by Arnett (2000), which is a stage of identity exploration, shifting social roles, and growing autonomy, as well as heightened vulnerability to mental disorders (Arnett et al., 2014; Tanner and Arnett, 2016). Depression and anxiety are especially common during this period. Roughly 31% of first-year students across eight countries screen positive for at least one common mental disorder (Auerbach et al., 2018). Furthermore, a nationwide survey of 41,620 students at 106 Chinese universities reported screen-positive rates of 9.8% for depression, 15.5% for anxiety, and 6.5% for both (Han et al., 2025). Moreover, these conditions consistently co-occur with impaired academic performance (Eisenberg et al., 2009), diminished quality of life, and suicidal ideation (Mortier et al., 2018). In this context, the rapid arrival of

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https://doi.org/10.1016/j.jad.2026.122428
Received 16 January 2026; Received in revised form 6 August 2026; Accepted 25 August 2026
Available online 27 August 2026
0165-0327/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

generative artificial intelligence (GenAI) has introduced a recent change in students' daily lives.

GenAI has quickly become part of young people's everyday lives, reshaping how they seek information, learn, and communicate. Its global user base had reached hundreds of millions by 2024, with young people forming the majority, whereas college students represent the heaviest adopters (Liu et al., 2024; Strzelecki, 2024). GenAI warrants particular attention because of its interactive nature. Unlike earlier digital tools, conversational GenAI engages users in a dialogue, responding immediately, adapting to the individual, generating human-like language, and remaining available around the clock, largely free of the interpersonal demands of human contact. These affordances are directly connected to emotional well-being via help-seeking, coping, self-disclosure, and social contact; hence, GenAI may serve as a source of support for some students and a focus of dependence for others. Still, it is unclear which pattern predominates and among whom.

Scholarly and policy attention has grown accordingly. Bibliometric analyses documented expanding research on AI and student mental health across fields of psychology, computer science, public health, and education (Chen et al., 2024; Liu et al., 2024). Additionally, the World Health Organization (2024) issued guidance on the ethics and governance of large multimodal models in health. However, this attention has not clarified how ordinary, non-clinical GenAI use relates to student mental health.

Evidence on this question is mixed. Several studies report that GenAI eases distress or supports emotional needs when delivered through purpose-built therapeutic applications, while other work ties heavier or escape-driven use to dependence, loneliness, and a weakened sense of belonging (Crawford et al., 2024; Heinz et al., 2025; Siddals et al., 2024; Yang et al., 2024; Yu et al., 2024). Part of this inconsistency likely stems from differences in the systems studied, the populations sampled, and how use itself was measured. It may also reflect something more specific: students probably engage with GenAI in qualitatively different ways, and these ways may carry different mental-health implications.

This possibility motivates a person-centered approach to studying GenAI use. Instead of estimating one overall association, we applied latent profile analysis to ask whether distinct, multidimensional patterns of use coexist within the same student population, and whether depression and anxiety differ across them (Laursen and Hoff, 2006).

1.1. Generative AI use and mental health: benefits and risks

Research on the mental-health implications of GenAI use points to both benefits and risks. On the benefit side, Siddals et al. (2024) found that users turning to GenAI chatbots for mental-health purposes described a sense of emotional refuge, useful guidance, and meaningful connection. In a clinical trial, Heinz et al. (2025) reported average symptom reductions of 51% among patients with major depressive disorder and 31% among those with generalized anxiety disorder using a purpose-built therapeutic chatbot. These systems were designed specifically for therapy rather than the general-purpose tools most students use, but the findings still suggest that GenAI may offer a supplementary channel of emotional support—particularly for those deterred from professional care by stigma or unable to access it due to limited resources (Wang et al., 2025).

Other work points in a different direction. Survey-based studies have tied heavier chatbot engagement to psychological dependence and greater loneliness or a diminished sense of belonging (Crawford et al., 2024; Yang et al., 2024). Students who rely on chatbots as a primary source of social support may withdraw from human contact at a cost to social connectedness and interpersonal skills (Chaudhry and Debi, 2024). Others frame excessive GenAI use as behavioral addiction: Yankouskaya et al. (2025) argued that the immediate, personalized, and continuously available responses of ChatGPT can foster dependency, leading users to form parasocial bonds that substitute for human relationships. Huang et al. (2024) found that mental health problems prospectively predicted later AI dependence among adolescents, with escape and social motivations mediating this pathway. Additionally, Yu et al. (2024) reported that escape-motivated use was positively associated with depression and anxiety. Therefore, motivation may be central to the link between GenAI use and mental health.

These findings imply that the relationship between GenAI use and mental health varies across users, and it may differ from patterns seen with earlier technologies. Excessive social-media use and smartphone addiction are associated with depression and anxiety (Chen et al., 2020; Elhai et al., 2017; Guo et al., 2023; Twenge et al., 2018), but GenAI provides human-like, real-time, quasi-social interaction that may generate distinct associations. Thus, a more systematic account must link specific usage patterns to specific outcomes.

1.2. Theoretical framework for understanding generative AI use and mental health

A person-centered approach like this can capture more than how frequently GenAI is used. Its mental-health correlates likely also depend on what people are seeking from GenAI, the skills they bring to using it, and the dispositions that shape how they engage. We organize these three dimensions—motivation, competence, and disposition—through complementary theoretical frameworks. First, Uses and Gratifications Theory (Katz et al., 1973) holds that people, as active media users, select technologies to satisfy various psychological and social needs—cognitive, affective, social-integrative, and tension release (Ruggiero, 2000). For GenAI, different motivations may carry different mental health implications. In a latent profile study of TikTok users, Ling et al. (2025) identified four motivation profiles, of which an "escapist addiction and novelty" profile showed the poorest mental health; escapism-driven social media use was likewise linked to a greater addiction risk and poorer well-being (Kardefelt-Winther, 2014). Further building on this framework, we distinguish four motivational dimensions—instrumental, entertainment, social, and escapism—to capture how patterns of need satisfaction may relate to different outcomes.

However, motivation alone does not determine the quality of engagement; cognitive resources also shape it. Consistent with the Unified Theory of Acceptance and Use of Technology (UTAUT; Venkatesh et al., 2003), which emphasizes users' expectations and facilitating conditions in technology adoption, we included AI literacy, defined as "the competencies that enable individuals to critically evaluate AI technologies, communicate and collaborate effectively with AI, and use AI as a tool" (Long and Magerko, 2020). Ng et al. (2021) characterized AI literacy as encompassing AI understanding, use, evaluation, and creation, as well as AI ethics. Higher AI literacy was linked to greater self-efficacy in AI use (Laupichler et al., 2022), whereas lower digital literacy was associated with a greater psychological burden during technology use (Bansal and Choudhary, 2024). Therefore, AI literacy may play a protective role in the relationship between AI use and mental health.

Beyond the questions of why and how, individual dispositions shape who is at the highest risk. Person-Environment Fit Theory (Edwards, 1991) postulates that the match between a person and an environment shapes adaptive outcomes. When applied to GenAI, users who differ in personality and emotion regulation may develop different usage patterns. A meta-analysis by Andreassen et al. (2012) found neuroticism positively correlated with social-media addiction in contrast to a negative correlation with conscientiousness. Wilson et al. (2010) reported similar links for conscientiousness. Specifically among AI chatbot users, Pentina et al. (2023) demonstrated that those with severe neuroticism were more likely to develop emotional dependency over time. Related self-regulatory factors showed comparable patterns: poor sleep was associated with more problematic technology use (Scott and Cleland Woods, 2018), with both representing a risk factor for and a consequence of depression and anxiety (Alvaro et al., 2013). Likewise,

executive-function deficits were implicated in problematic technology use (Oh et al., 2021). Thus, highly conscientious users may gravitate toward goal-directed use, whereas those with severe neuroticism or self-regulation difficulties may be more prone to escapist or problematic patterns.

Taken together, these three frameworks sketch an integrated picture of why GenAI use might be heterogeneous: motivation captures what users are seeking, competence shapes how they evaluate and use the technology, and disposition affects how well they regulate their engagement with it. These dimensions probably interact rather than operate in isolation—low AI literacy, for example, may push users toward escapist rather than instrumental use, and conscientiousness may influence both why students turn to GenAI in the first place and how well they keep that use in check. The question this raises is not whether GenAI is uniformly beneficial or harmful, but which configurations of use go together with different mental-health outcomes.

1.3. Variable-centered versus person-centered research perspectives

How that question gets answered depends partly on the analytic lens applied. Most existing research relies on variable-centered strategies that estimate an average association between usage frequency or duration and mental health. Because these strategies focus on population-average effects, they can mask real heterogeneity among individuals (Laursen and Hoff, 2006), and they typically examine one dimension of use at a time rather than how behavior, motivation, AI literacy, and dependency combine into distinct patterns. An overall association can therefore look weak, inconsistent, or sample-dependent even when meaningful subgroup patterns are present underneath it.

A person-centered approach sidesteps this problem. Latent profile analysis (LPA) groups individuals into subgroups that share similar patterns across several observed indicators, making population heterogeneity visible rather than averaging it away (Nylund et al., 2007). It has been widely applied to technology use and mental health: studies using large adolescent samples identified profiles of social media and internet addiction that differ in risk and depression and anxiety levels (Bányai et al., 2017; Cheng et al., 2022; Wang et al., 2024). However, person-centered evidence on GenAI use remains scarce, with the few existing studies profiling a single dimension, such as students' self-reporting reliance on ChatGPT (Stojanov et al., 2024), rather than the joint configuration of usage behavior, motivation, AI literacy, and dependency. Thus, identifying user subgroups from these multidimensional characteristics and examining how each relates to depression and anxiety are important next steps. Such evidence cannot establish whether GenAI benefits one group and harms another, but it can test whether mixed aggregate findings are consistent with population heterogeneity and identify configurations that warrant longitudinal research.

1.4. The present study

Accordingly, the present study used LPA in a large sample of Chinese college students to identify heterogeneous subgroups of GenAI use across behavior, motivation, AI literacy, and dependency dimensions and examine whether these configurations differed systematically in relation to depression and anxiety. Therefore, we treated the mixed literature as a possible reflection of user heterogeneity rather than an unresolved contradiction: divergent findings may arise when studies sample different mixtures of users whose configurations of use carry more or less favorable mental health correlates. Then, a complementary machine-learning analysis examined which demographic and psychological characteristics (including personality traits, emotion-regulation strategies, and sleep quality) most clearly distinguished these profiles, thereby identifying correlates for further longitudinal research.

2. Methods

2.1. Participants

Participants were recruited from four universities in western China, including comprehensive universities, universities of science and technology, and normal universities. A multistage, class-based strategy was used: universities were selected to cover these common institution types, followed by student recruitment through participating classes. The study population comprised full-time undergraduate and graduate students who had used at least one GenAI tool during the preceding three months. Inclusion criteria were (1) current enrollment in a full-time undergraduate or graduate program; (2) use of at least one GenAI tool in the preceding three months; (3) the ability to understand and complete the questionnaire independently; and (4) voluntary participation with informed consent. Exclusion criteria were current psychotherapy or psychotropic medication use, a serious physical illness that substantially impaired academic or daily functioning, or a major life event in the preceding six months. Applying these criteria kept the analytic sample restricted to the general, non-clinical student population, so that the associations of interest would not be driven by active clinical treatment, substantial medical burden, or recent acute stressors.

In total, 5923 students completed and submitted the survey. Eligibility screening excluded 142 respondents who failed to meet at least one inclusion criterion, and a further 33 responses were dropped during data-quality screening because of implausibly short completion times or invariant response patterns—both commonly used markers of insufficient effort responding (Meade and Craig, 2012). Thus, the final analytic sample comprised 5748 participants, representing 97.05% of all submitted surveys. Although this sample was large, accurate class enumeration in LPA depends not on the sample size alone but also on profile separation, the number of indicators, relative profile sizes, and model specification. Therefore, the retained solution was selected using multiple statistical and substantive criteria rather than only the sample size (Nylund et al., 2007; Tein et al., 2013). The study was conducted according to the ethical principles of the Declaration of Helsinki and approved by the institutional Ethics Committee for Scientific Research, School of Psychology, Northwest Normal University.

2.2. Measures

2.2.1. Demographic information

Demographic information was collected with a researcher-developed questionnaire covering age, gender, academic discipline, place of origin (urban vs. rural), only-child status, and body mass index (BMI). Family socioeconomic status (SES) was assessed by the four-dimensional scale of Zhao et al. (2022), evaluating material wealth, per-capita monthly household income, parental education, and parental occupational status (Supplementary Methods), with the total scores ranging from 4 to 20. In the present sample, Cronbach's α for this scale was 0.87.

2.2.2. Generative AI use assessment

Generative AI use was assessed across several dimensions.

2.2.2.1. AI use behaviors. Two researcher-developed items assessed usage frequency (on a 5-point scale) and average daily use duration (in hours).

2.2.2.2. AI usage motivation. Motivation for AI use was measured via the AI Usage Motivation Scale revised by Huang et al. (2024). Grounded in the uses-and-gratifications framework, the scale comprises 12 items across four dimensions—instrumental, entertainment, social, and escapism motivation—each rated on a 4-point Likert scale. Cronbach's α was 0.88, and confirmatory factor analysis supported the four-factor structure.

2.2.2.3. AI literacy. AI literacy was measured using the Artificial Intelligence Literacy Scale for Chinese College Students (AILS-CCS; Ma and Chen, 2024), which comprised 15 items (rated on a 5-point Likert scale) across four dimensions: cognitive, usage, evaluative, and ethical. Cronbach's α was 0.91, and confirmatory factor analysis supported the four-factor structure.

2.2.2.4. AI dependency. AI dependency was measured through the five-item AI Dependency Scale, adapted by Huang et al. (2024) from validated smartphone-addiction measures within the components model of behavioral addiction to capture over-reliance, excessive time, relationship impairment, withdrawal, and loss of control. Items use a 4-point Likert scale. In the present sample, the scale was clearly unidimensional, with good model fit (CFI = 0.995, TLI = 0.987, SRMR = 0.018) and strong reliability (ω = 0.94, Cronbach's α = 0.89), and its average variance extracted (AVE = 0.74) indicated adequate convergent validity; full psychometric details appear in Supplementary Tables S2 and S3.

2.2.3. Depression, anxiety, and sleep quality assessment

Depressive symptoms were measured with the Chinese version of the Beck Depression Inventory–II (BDI-II; Wang et al., 2011), a 21-item scale with total scores from 0 to 63 (Cronbach's α = 0.92). Anxiety symptoms were assessed with the Chinese version of the Beck Anxiety Inventory (BAI; Liang et al., 2018), also 21 items, with total scores from 0 to 63 (α = 0.93). Sleep quality was assessed with the Chinese version of the Pittsburgh Sleep Quality Index (PSQI; Liu et al., 2005); total scores range from 0 to 21, with higher scores indicating poorer sleep (α = 0.83).

2.2.4. Other psychological measures

2.2.4.1. Emotion regulation. Emotion regulation was assessed through the Chinese version of the Emotion Regulation Questionnaire (ERQ; Gross and John, 2003), a 10-item measure of cognitive reappraisal and expressive suppression rated on a 7-point Likert scale. Cronbach's α was 0.86 for reappraisal and 0.79 for suppression.

2.2.4.2. Executive function. Executive function was evaluated by the Chinese version of the Executive Function Index–Short Form (EFI-S; Spinella, 2005), comprising 14 items rated on a 5-point scale, with higher scores reflecting greater executive-function difficulties. Cronbach's α was 0.88.

2.2.4.3. Smartphone addiction. Smartphone addiction was assessed via the Chinese version of the Mobile Phone Addiction Index (MPAI; Leung, 2008), a 17-item measure rated on a 5-point Likert scale (α = 0.90).

2.2.4.4. Personality traits. The five broad personality traits—extraversion, agreeableness, conscientiousness, neuroticism, and openness—were evaluated through the Chinese version of the Big Five Inventory–Short Form (BFI-S; Zhang et al., 2019), which is a 15-item measure rated on a 5-point Likert scale. Across the five subscales, Cronbach's α ranged from 0.74 to 0.82.

2.3. Procedure

Data were collected between September and November 2025. The survey was administered to participating classes under standardized instructions. Students who agreed to participate provided electronic informed consent and completed the questionnaire online in a computer laboratory, supervised by trained psychology graduate students. Completion took approximately 30 min; afterward, participants received a small token of appreciation.

2.4. Data analysis

2.4.1. Preliminary analyses

Missing data were minimal (<5% for all variables) and were handled with full-information maximum likelihood estimation in the latent profile models (Schafer and Graham, 2002). We examined bivariate associations among the study variables with Pearson correlations and applied the Benjamini-Hochberg procedure to control the false discovery rate (Benjamini and Hochberg, 1995). Preliminary analyses were run in SPSS 27.0 and R 4.3.2.

2.4.2. Latent profile analysis

We conducted LPA in R (version 4.3.2) with the tidyLPA package to identify heterogeneous patterns of generative AI use. Based on theory and prior research, we entered eight indicators: AI usage frequency, daily usage duration, four usage motivations (instrumental, entertainment, social, and escapism), total AI literacy score, and AI dependency—together spanning the behavioral, motivational, competency, and dependency dimensions of AI use. All indicators were standardized (z-scored) beforehand so that they would be comparable across measurement scales.

We estimated models specifying one to six profiles sequentially, with parameters estimated using the mclust package (Scrucca et al., 2016) under equal, diagonal variance-covariance matrices across profiles (the EEI model), which assumes local independence of indicators within profiles. We chose this parsimonious, class-invariant covariance structure—the most restricted of the available Gaussian-mixture parameterizations and the default in widely used latent profile software—to avoid over-parameterizing a model with eight standardized indicators and to promote stable convergence across candidate solutions (Masyn, 2013; Pastor et al., 2007; Spurk et al., 2020; Vermunt and Magidson, 2002).

The number of profiles was determined from a combination of criteria. First, we examined information criteria AIC, BIC, and sample-size-adjusted BIC (aBIC), with lower values indicating better fit. The BIC weighted most heavily, as it is the most reliable index for class enumeration (Nylund et al., 2007). Second, we assessed classification precision using entropy, which ranges from 0 to 1, with higher values indicating more precise classification (Celeux and Soromenho, 1996). Third, we conducted likelihood-ratio tests: the Lo-Mendell-Rubin adjusted LRT (LMR-LRT; Lo et al., 2001) and the bootstrap LRT (BLRT; McLachlan and Peel, 2000); for them, a significant p-value (< 0.05) indicates that a k-profile model fits better than a $(k - 1)$-profile model. Fourth, we required each profile to comprise at least 5% of the sample (Lubke and Muthén, 2005). Fifth, we needed the retained solution to be theoretically meaningful and practically interpretable. The LMR-LRT and BLRT were computed in Mplus (version 8.10; Muthén & Muthén, 1998–2017), which default class-invariant, diagonal covariance specification corresponds to the EEI model estimated in tidyLPA/mclust.

Once the number of profiles was established, differences across profiles were examined using one-way analyses of variance for continuous variables and chi-square tests for categorical variables, with Tukey's HSD post-hoc comparisons corrected for multiple testing using the Benjamini-Hochberg procedure (Benjamini and Hochberg, 1995). We additionally conducted analyses of covariance controlling for gender, age, and family SES, as well as a multinomial logistic regression predicting profile membership from these covariates, to confirm that the between-profile differences in mental health were not driven by demographic composition.

2.4.3. Machine-learning classification

We conducted an exploratory random forest classification (Breiman, 2001), implemented in Python (version 3.10.11) with scikit-learn (version 1.3.0; Pedregosa et al., 2011) and Optuna (version 3.5.0; Akiba et al., 2019) to further characterize the profiles, which served as

the classification target. Predictors comprised depression, anxiety, and sleep quality; executive function, smartphone addiction, emotion-regulation strategies (cognitive reappraisal and expressive suppression), the Big Five personality traits, age, family SES, and BMI. The AI use indicators that defined the latent profiles were excluded from the predictor set; hence, the profiles were predicted from psychological and demographic correlates that were not used to define them.

The data were divided by a stratified 70/30 split into a training set of 4023 cases and an independent held-out test set of 1725 cases, applying stratification on profile membership to preserve class proportions in each set (details in the Supplementary Methods). We performed nested (double) cross-validation using the training set to select hyperparameters while avoiding the optimistic bias arising when the same data are used for tuning and evaluation (Varma and Simon, 2006). Within each outer fold (stratified, $k = 5$), an inner loop (stratified, $k = 3$) tuned hyperparameters via the Optuna framework, with performance evaluated in the outer loop to reduce optimism arising from model selection (Zhang et al., 2026). Then, the final model was trained on the entire training set using the selected hyperparameters and evaluated on the independent held-out test set, deriving all reported performance metrics from the latter. Classification performance was summarized using accuracy, balanced accuracy, per-class precision/recall/F_1, macro-F_1, and one-versus-rest area under the curve (AUC). Uncertainty intervals for AUC and pairwise AUC differences were obtained from 3000 bootstrap resamples of the held-out test cases. Feature contributions were quantified using SHAP values (Lundberg and Lee, 2017; Lundberg et al., 2020).

3. Results

3.1. Sociodemographic characteristics and correlations

The final analytic sample consisted of 5748 Chinese college students; sociodemographic characteristics and descriptive statistics for all study variables appear in Supplementary Table S1. Participants averaged 20.73 years old (standard deviation [*SD*] = 2.51), and roughly two-thirds were female (67.86%). For mental health, 15.41% reported moderate or greater depressive symptoms and 17.30% reported moderate or greater anxiety.

Table 1 presents the bivariate correlations among the study variables. Usage frequency and daily duration were moderately correlated ($r = 0.54$, $p_{FDR} < 0.001$), and both related positively to all four use motivations, most strongly to escapism. AI literacy was positively associated with instrumental motivation ($r = 0.39$, $p_{FDR} < 0.001$) but negatively associated with escapism motivation ($r = -0.21$, $p_{FDR} < 0.001$). AI dependency showed its strongest association with escapism motivation ($r = 0.58$, $p_{FDR} < 0.001$) and a negative association with AI literacy ($r = -0.24$, $p_{FDR} < 0.001$).

Correlations with the mental health variables showed a coherent pattern. Daily duration correlated more strongly with depression ($r = 0.28$) and anxiety ($r = 0.24$) in contrast to usage frequency (both $p_{FDR} < 0.001$). Among the four motivations, escapism was most closely linked to depression ($r = 0.41$) and anxiety ($r = 0.36$), whereas instrumental motivation showed weak negative or nonsignificant associations with both. AI literacy correlated negatively, although modestly, with depression ($r = -0.19$) and anxiety ($r = -0.16$). AI dependency correlated positively and moderately with depression, anxiety, and smartphone addiction (all $p_{FDR} < 0.001$).

3.2. Latent profile analysis

3.2.1. Model selection

We performed LPA using eight indicators—usage frequency, daily duration, the four use motivations (instrumental, entertainment, social, and escapism), total AI literacy score, and dependency—and estimated solutions with one to six profiles sequentially. Fit indices for all models

Table 1
Pearson correlation matrix of AI use variables and mental health indicators.

Variable	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1. Usage Frequency	–														
2. Daily Usage Duration	0.54***	–													
3. Instrumental Motivation	0.29***	0.21***	–												
4. Entertainment Motivation	0.35***	0.41***	0.08*	–											
5. Social Motivation	0.31***	0.36***	0.11**	0.49***	–										
6. Escapism Motivation	0.38***	0.44***	−0.04	0.53***	0.45***	–									
7. AI Literacy	0.18***	0.06	0.39***	−0.11**	−0.07*	−0.21***	–								
8. Dependency Level	0.49***	0.55***	0.08*	0.46***	0.41***	0.58***	−0.24***	–							
9. Depression	0.21***	0.28***	−0.08*	0.32***	0.27***	0.41***	−0.19***	0.38***	–						
10. Anxiety	0.18***	0.24***	−0.05	0.29***	0.24***	0.36***	−0.16***	0.34***	0.62***	–					
11. Sleep Quality	0.16***	0.23***	−0.04	0.24***	0.19***	0.31***	−0.13***	0.28***	0.46***	0.42***	–				
12. Executive Function	0.15***	0.21***	−0.11**	0.22***	0.18***	0.27***	−0.18***	0.26***	0.39***	0.34***	0.32***	–			
13. Smartphone Addiction	0.36***	0.43***	0.02	0.38***	0.33***	0.45***	−0.15***	0.49***	0.36***	0.32***	0.28***	0.31***	–		
14. Cognitive Reappraisal	−0.09**	−0.12***	0.19***	−0.14***	−0.11**	−0.21***	0.26***	−0.18***	−0.26***	−0.22***	−0.18***	−0.14***	−0.12***	–	
15. Expressive Suppression	0.11**	0.15***	−0.01	0.18***	0.16***	0.22***	−0.08*	0.21***	0.22***	0.18***	0.15***	0.13***	0.17***	−0.06	–

Note. Correlations are Pearson coefficients; *p*-values were adjusted with the Benjamini-Hochberg procedure.
* $p < .05$.
** $p < .01$.
*** $p < .001$.

are reported in Table 2 and Fig. 1A.

Several converging indices favored the four-profile solution. It had the lowest BIC (BIC = 102,970.17), an improvement over the three-profile model (ΔBIC = 835.06), before BIC rose again at five profiles (BIC = 103,147.91). The AIC kept declining as profiles were added, but BIC is generally considered the more trustworthy index for class enumeration (Nylund et al., 2007). Entropy for the four-profile model was 0.81, indicating adequate classification precision, and both the LMR-LRT and BLRT favored four profiles over three ($p < .01$). At five profiles, the LMR-LRT was no longer significant, and although the BLRT remained significant ($p < .05$), its likelihood-ratio increment was much smaller; at six profiles, neither test was significant. Finally, all four profiles were substantively interpretable and of adequate size—even the smallest (Problem-Dependent), at 15.0% of the sample, comfortably cleared the conventional 5% threshold. Weighing statistical fit, class size, and interpretability together, we retained the four-profile solution.

3.2.2. Profile characteristics

Under the retained solution, the 5748 students fell into four profiles. Fig. 1B displays standardized indicator means, and Table 3 reports the corresponding values and between-profile comparisons. Rational-Tool users (n = 1609, 27.99%) formed a high-literacy, low-dependency, task-oriented group: moderately low on frequency and duration, highest on instrumental motivation, lowest on the remaining motivations, highest on AI literacy, and lowest on dependency among the four profiles. Moderate-Recreational users (n = 1954, 33.99%)—the largest group—showed moderate frequency, duration, literacy, and dependency, distinguished chiefly by comparatively elevated entertainment and social motivation. Problem-Dependent users (n = 862, 15.00%)—the smallest group—constituted the clearest high-risk profile, reporting the highest frequency, duration, and escapism motivation; the lowest instrumental motivation and AI literacy, and markedly elevated dependency. Light-Exploratory users (n = 1323, 23.02%) were minimally engaged, showing the lowest frequency and duration of all profiles and uniformly low motivation and dependency.

3.2.3. Between-profile differences

Table 3 provides between-profile differences in sociodemographic characteristics, mental health indicators, and other psychological variables.

3.2.3.1. Sociodemographic characteristics. Rational-Tool users were the oldest and significantly older than Problem-Dependent users ($F = 7.84$, $p < .001$). Gender composition differed across the profiles ($\chi^2 = 31.24$, $p < .001$), with women most heavily represented in the Problem-Dependent profile (75.68%). Likewise, family SES differed ($F = 38.92$, $p < .001$), being the highest among Rational-Tool users and the lowest among Problem-Dependent users.

3.2.3.2. Mental-health indicators. The profiles differed in depressive symptoms ($F = 142.67$, $p < .001$, $\eta^2 = 0.07$), with Problem-Dependent users scoring the highest (M = 16.87, in the mild range) and Rational-Tool users scoring the lowest (M = 8.14, in the minimal range; Fig. 1C). Anxiety ($F = 124.58$, $p < .001$, $\eta^2 = 0.06$) and sleep quality ($F = 108.34$, $p < .001$, $\eta^2 = 0.05$) followed the same pattern.

3.2.3.3. Executive function and behavioral addiction. Problem-Dependent users scored the highest for executive-function deficits ($F = 134.21$, $p < .001$, $\eta^2 = 0.07$) and smartphone addiction ($F = 156.78$, $p < .001$, $\eta^2 = 0.08$), whereas Rational-Tool users scored the lowest for both.

3.2.3.4. Emotion regulation and personality. Cognitive reappraisal ($F = 87.45$, $p < .001$, $\eta^2 = 0.04$) followed the order of Rational-Tool > Light-Exploratory > Moderate-Recreational > Problem-Dependent, with expressive suppression showing the reverse ordering. The most pronounced differences emerged for neuroticism ($F = 124.87$, $p < .001$, $\eta^2 = 0.06$) and conscientiousness ($F = 118.47$, $p < .001$, $\eta^2 = 0.06$). Problem-Dependent users showed the highest levels of neuroticism and the lowest levels of conscientiousness, whereas Rational-Tool users showed the opposite pattern.

3.3. Machine-learning classification

We conducted an exploratory random forest classification to assess how well the four profiles could be recovered from psychological and demographic correlates not used to define them and identify the correlates that best distinguished them. On the held-out test set, the classifier reached an overall accuracy of 0.810, a balanced accuracy of 0.808, and a macro-averaged F_1 of 0.818. The macro one-versus-rest AUC—computed as the area under the pointwise mean of the four interpolated one-versus-rest receiver operating characteristic (ROC) curves—was 0.894 (95% confidence interval [CI]: 0.883, 0.904; Fig. 2A). Discrimination was the strongest for the Problem-Dependent profile (AUC = 0.938 [0.919, 0.954]) and the Rational-Tool profile (AUC = 0.893 [0.874, 0.910]) and somewhat lower for the Moderate-Recreational (AUC = 0.873 [0.853, 0.890]) and Light-Exploratory (AUC = 0.872 [0.852, 0.890]) profiles. Per-class precision, recall, and F_1 were 0.784/0.797/0.791 for Rational-Tool, respectively, 0.777/0.869/0.820 for Moderate-Recreational, respectively, 0.949/0.857/0.901 for Problem-Dependent, respectively, and 0.817/0.710/0.760 for Light-Exploratory, respectively.

Because the four profiles differed in size, we examined whether classification was shaped by class proportions. Recall was at least 0.71 for every profile. Despite being the smallest class, the Problem-Dependent profile achieved high recall (0.857), indicating that its strong AUC was not accompanied by poor minority-class detection. Misclassifications occurred mainly among the three lower-risk profiles, with the Light-Exploratory profile most often confused with the Rational-Tool and Moderate-Recreational profiles (Fig. 2B). Paired bootstrap comparisons (3000 resamples) confirmed that the Problem-Dependent AUC exceeded those of the other three profiles (ΔAUC = 0.045–0.066, all 95% CIs excluding zero; Supplementary Fig. S3),

Table 2
Model fit indices for the latent profile analysis of generative AI use patterns.

Number of profiles	*LL*	AIC	BIC	aBIC	Entropy	LMR-LRT	BLRT
1	−54,123.47	108,278.94	108,385.21	108,325.87	–	–	–
2	−52,487.63	105,031.26	105,218.73	105,115.42	0.71	3241.18***	3271.68***
3	−51,734.28	103,536.56	103,805.23	103,658.95	0.74	1486.92***	1506.70***
4	**−51,267.15**	**102,620.30**	**102,970.17**	**102,780.92**	**0.81**	**921.34*** **	**934.26*** **
5	−51,089.42	102,272.84	103,147.91	102,871.69	0.76	347.82**	355.46*
6	−50,978.31	102,058.62	103,258.89	102,895.70	0.73	216.54	222.22

Note. LL, log-likelihood; AIC, Akaike information criterion; BIC, Bayesian information criterion; aBIC, sample-size-adjusted BIC; LMR-LRT, Lo-Mendell-Rubin–adjusted likelihood-ratio test; BLRT, bootstrap likelihood-ratio test (5000 resamples). The retained four-profile solution is shown in bold.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

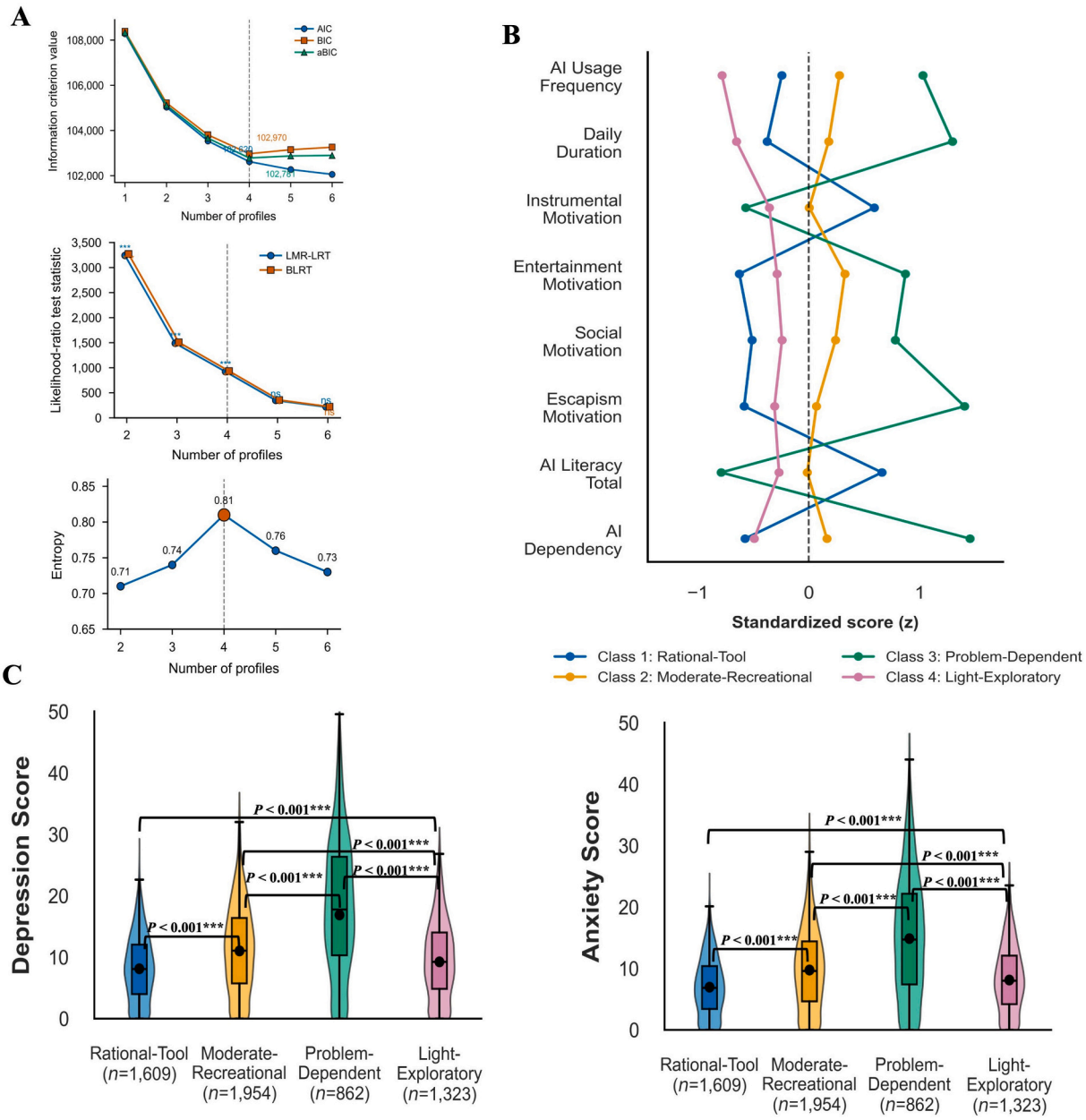


Fig. 1. Latent-profile model selection, generative-AI use patterns, and mental health outcomes.

Note. (A) Model-selection indices for the one- through six-profile solutions. From top to bottom: the Akaike information criterion (AIC), Bayesian information criterion (BIC), and sample-size-adjusted BIC (aBIC), for which lower values indicate better relative fit; the Lo-Mendell-Rubin–adjusted likelihood-ratio test (LMR-LRT) and bootstrap likelihood-ratio test (BLRT); and entropy, for which higher values indicate greater classification precision. Dashed vertical lines mark the retained four-profile solution. (B) Weighted standardized mean scores (*z* scores) for the eight profile indicators across the Rational-Tool, Moderate-Recreational, Problem-Dependent, and Light-Exploratory profiles; the vertical dashed line denotes $z = 0$. (C) Distributions of depression (left) and anxiety (right) across the Rational-Tool ($n = 1609$), Moderate-Recreational ($n = 1954$), Problem-Dependent ($n = 862$), and Light-Exploratory ($n = 1323$) profiles. Violin widths represent kernel-density estimates; boxes span the interquartile range, central lines indicate medians, whiskers extend to the most extreme observations within 1.5 × the interquartile range, and black circles indicate means. $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

representing an advantage that pertains to ranking-based discrimination (AUC) rather than to threshold-dependent classification.

To test whether this advantage arose from the training-set imbalance, we re-trained the classifier under random undersampling and, separately, SMOTE oversampling (Chawla et al., 2002), which was applied to the training data only and evaluated on the same unresampled test set. Overall and profile-specific performance were largely unchanged, with the Problem-Dependent AUC stable across both schemes (0.932 and 0.936, vs. 0.938) and the expected precision-recall trade-off under resampling (Supplementary Tables S5 and S6). Thus, the discriminative advantage of the profile did not stem from the original class distribution.

Afterward, SHAP analysis examined the contribution of each correlate to classification (Fig. 2C and D). Rankings were based on class-specific mean absolute SHAP values, with each beeswarm panel displaying the ten highest-ranking features for that class. Conscientiousness contributed most to the Rational-Tool profile, followed by executive function and smartphone addiction. Smartphone addiction contributed most to the Moderate-Recreational profile, followed by executive function and depression. Depression contributed most to the Problem-Dependent profile, followed by smartphone addiction and executive-function difficulties, indicating together with the group comparisons

Table 3
Comparisons of sociodemographic, AI use, and psychological variables across the four AI user profiles.*

Variable	Rational-Tool (n = 1609)	Moderate-Recreational (n = 1954)	Problem-Dependent (n = 862)	Light-Exploratory (n = 1323)	F/χ^2	η^2/V	Post-hoc
Sociodemographics							
Age (years)	21.02 ± 2.47	20.81 ± 2.53	20.34 ± 2.71	20.58 ± 2.44	7.84***	0.004	1 > 3
Gender					31.24***	0.07	–
Male	36.22%	30.87%	24.32%	33.41%			
Female	63.78%	69.13%	75.68%	66.59%			
Academic Discipline					19.87**	0.04	–
Humanities	27.18%	31.54%	34.12%	29.21%			
Engineering	37.86%	33.19%	30.58%	35.67%			
Science	34.96%	35.27%	35.30%	35.12%			
Only Child					14.67**	0.05	–
Yes	40.14%	37.23%	31.64%	38.08%			
No	59.86%	62.77%	68.36%	61.92%			
Place of Origin					22.41***	0.06	–
Urban	73.51%	68.04%	61.28%	69.78%			
Rural	26.49%	31.96%	38.72%	30.22%			
Family SES	12.04 ± 3.56	10.97 ± 3.71	9.84 ± 3.98	11.21 ± 3.61	38.92***	0.02	1 > 2, 4 > 3
BMI (kg/m^2)	21.34 ± 3.18	21.47 ± 3.29	21.58 ± 3.41	21.38 ± 3.24	0.89	0.00	–
AI use characteristics[a]							
Usage frequency (1–5)	2.87 ± 0.81	3.38 ± 0.87	4.12 ± 0.64	2.34 ± 0.76	487.24***	0.20	3 > 2 > 1 > 4
Daily duration (hours)	1.24 ± 0.71	1.98 ± 1.14	3.47 ± 1.68	0.87 ± 0.54	521.38***	0.21	3 > 2 > 1 > 4
Usage Motivation							
Instrumental (1–4)	3.58 ± 0.52	3.14 ± 0.68	2.71 ± 0.84	2.87 ± 0.74	142.18***	0.07	1 > 2 > 4, 3
Entertainment (1–4)	1.87 ± 0.72	2.74 ± 0.81	3.24 ± 0.71	2.18 ± 0.84	298.56***	0.13	3 > 2 > 4 > 1
Social (1–4)	1.74 ± 0.68	2.41 ± 0.87	2.89 ± 0.82	1.98 ± 0.81	198.42***	0.09	3 > 2 > 4 > 1
Escapism (1–4)	1.52 ± 0.64	2.14 ± 0.81	3.41 ± 0.67	1.78 ± 0.74	512.34***	0.21	3 > 2 > 4 > 1
AI Literacy							
Cognitive	3.97 ± 0.68	3.42 ± 0.81	2.74 ± 0.94	3.18 ± 0.87	201.45***	0.10	1 > 2 > 4 > 3
Usage	3.84 ± 0.71	3.31 ± 0.88	2.58 ± 0.97	3.08 ± 0.91	187.32***	0.09	1 > 2 > 4 > 3
Evaluative	3.71 ± 0.74	3.08 ± 0.86	2.47 ± 0.92	2.91 ± 0.88	178.56***	0.09	1 > 2 > 4 > 3
Ethical	3.58 ± 0.84	2.97 ± 0.98	2.31 ± 1.02	2.74 ± 0.97	154.23***	0.07	1 > 2 > 4 > 3
Total score (1–5)	3.78 ± 0.62	3.20 ± 0.76	2.53 ± 0.87	2.98 ± 0.82	267.89***	0.12	1 > 2 > 4 > 3
Dependency (1–4)	1.78 ± 0.71	2.58 ± 0.87	3.98 ± 0.74	1.87 ± 0.78	578.42***	0.23	3 > 2 > 1, 4
Mental health							
Depression (BDI-II)	8.14 ± 5.87	11.02 ± 8.14	16.87 ± 11.54	9.24 ± 6.78	142.67***	0.07	3 > 2 > 4, 1
Anxiety (BAI)	6.98 ± 5.12	9.74 ± 7.21	14.87 ± 10.24	8.12 ± 5.89	124.58***	0.06	3 > 2 > 4, 1
Sleep quality (PSQI)	5.54 ± 2.81	6.87 ± 3.42	9.18 ± 4.14	6.14 ± 3.21	108.34***	0.05	3 > 2 > 4, 1
Other psychological variables							
Executive function (EFI-S)	31.24 ± 8.14	35.47 ± 10.24	43.87 ± 13.47	33.18 ± 9.24	134.21***	0.07	3 > 2 > 4 > 1
Smartphone addiction (MPAI)	35.14 ± 10.87	42.18 ± 13.14	52.47 ± 16.87	38.24 ± 11.87	156.78***	0.08	3 > 2 > 4 > 1
Emotion Regulation (ERQ)							
Cognitive reappraisal	4.91 ± 1.18	4.42 ± 1.31	3.58 ± 1.47	4.67 ± 1.24	87.45***	0.04	1 > 4 > 2 > 3
Expressive suppression	3.78 ± 1.21	4.14 ± 1.28	4.87 ± 1.41	3.91 ± 1.24	64.32***	0.03	3 > 2, 4 > 1
Personality (BFI-S)							
Extraversion	3.41 ± 0.87	3.14 ± 0.94	2.78 ± 1.04	3.24 ± 0.91	42.18***	0.02	1 > 4, 2 > 3
Agreeableness	3.71 ± 0.74	3.58 ± 0.78	3.47 ± 0.84	3.64 ± 0.77	8.42***	0.004	1 > 3
Conscientiousness	3.74 ± 0.78	3.28 ± 0.87	2.71 ± 0.98	3.24 ± 0.89	118.47***	0.06	1 > 2, 4 > 3
Neuroticism	2.68 ± 0.84	3.04 ± 0.91	3.71 ± 0.97	2.84 ± 0.88	124.87***	0.06	3 > 2 > 4, 1
Openness	3.54 ± 0.78	3.37 ± 0.82	3.24 ± 0.87	3.41 ± 0.79	12.47***	0.006	1 > 3

Note. F, one-way ANOVA (continuous variables); χ^2, chi-square test (categorical variables); post-hoc comparisons used Tukey's HSD ($p < .05$). Primary outcomes (depression and anxiety) are shown in bold. Profiles: 1 = Rational-Tool, 2 = Moderate-Recreational, 3 = Problem-Dependent, 4 = Light-Exploratory. SES, socio-economic status; BDI-II, Beck Depression Inventory–II; BAI, Beck Anxiety Inventory; PSQI, Pittsburgh Sleep Quality Index; EFI-S, Executive Function Index–Short Form; MPAI, Mobile Phone Addiction Index; ERQ, Emotion Regulation Questionnaire; BFI-S, Big Five Inventory–Short Form.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

[a] These variables served as the profile indicators in the LPA; large between-profile effect sizes are therefore expected since LPA maximizes separation on the defining indicators.

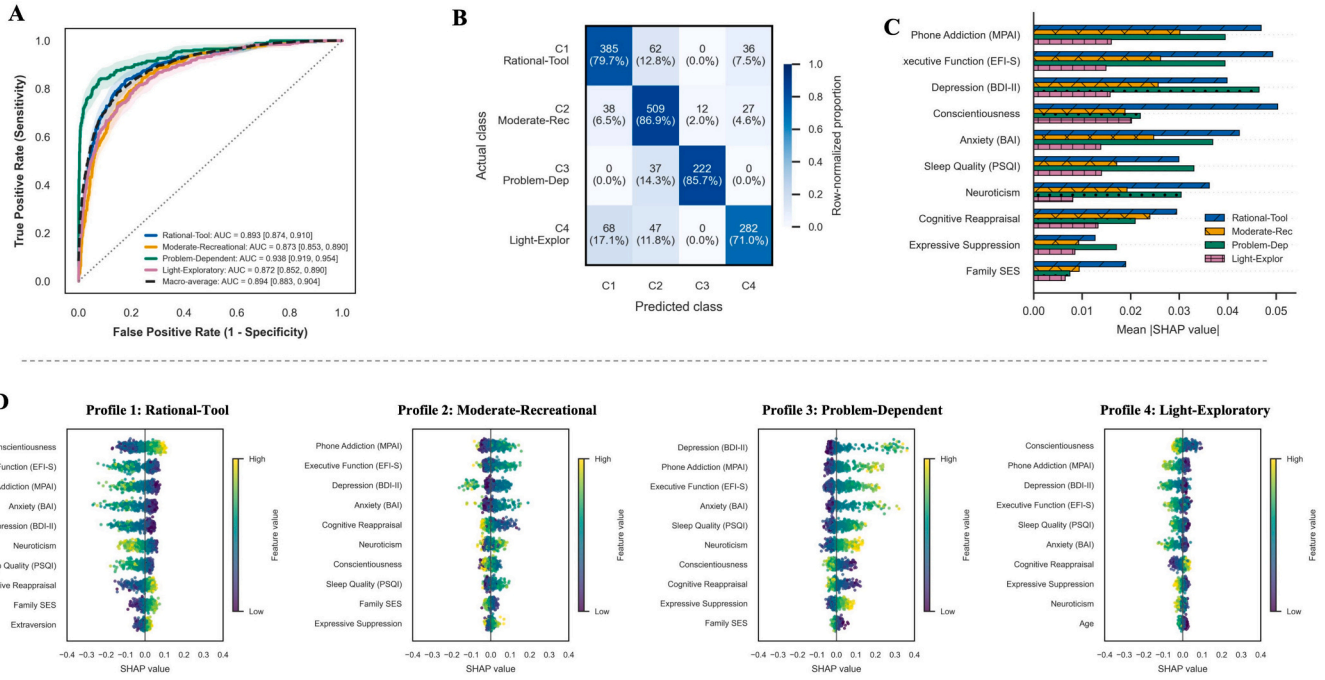


Fig. 2. Random-forest classification performance and SHAP feature contributions for the four AI use profiles.
Note. (A) One-versus-rest receiver operating characteristic (ROC) curves evaluated on a stratified 30% holdout test set (n = 1725) with bootstrap 95% confidence intervals. The dashed curve represents the macro-average ROC, and the diagonal denotes chance-level performance. (B) Row-normalized confusion matrix showing the percentage of observations in each actual class assigned to each predicted class. (C) Class-specific mean absolute SHAP values for the ten most globally important features, with global importance defined as the mean absolute SHAP value averaged across the four classes. (D) Class-specific SHAP beeswarm plots for the Rational-Tool, Moderate-Recreational, Problem-Dependent, and Light-Exploratory profiles. Each panel displays the ten highest-ranking features for that class; consequently, the features shown and their ordering may differ across the panels. Each point represents one sampled test-set observation. Its horizontal position indicates the SHAP value, with positive and negative values respectively increasing and decreasing the model output for the corresponding class. Point color denotes the within-feature normalized value (dark green = low; yellow = high). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

that this profile co-occurred with greater depressive symptoms, broader technology dependence, and executive-function difficulties. Conscientiousness again contributed most to the Light-Exploratory profile, followed by smartphone addiction and depression. Because depression and anxiety were themselves among the outcomes distinguishing the profiles, their contributions here should be read as confirmatory associations rather than as independent or prospective predictors.

As a sensitivity analysis that can address this overlap, we refit the classifier after removing depression (BDI-II) and anxiety (BAI) from the predictor set, using a held-out test set and the same modeling pipeline. Threshold-based performance declined relative to the primary model (accuracy: 0.810 to 0.689; balanced accuracy: 0.808 to 0.679; macro-F_1: 0.818 to 0.690), whereas macro-AUC discrimination remained good but decreased from 0.894 to 0.876 [0.861, 0.887]. The Problem-Dependent profile retained the highest one-versus-rest AUC (0.943 [0.928, 0.954]), but its recall was 0.614. Thus, high AUC discrimination did not imply uniformly strong threshold-based classification (Supplementary Fig. S2 and Supplementary Table S5). SHAP plots continued to emphasize conscientiousness, phone addiction, and executive function across profiles, whereas neuroticism was among the top three for the Problem-Dependent profile after symptom removal. Therefore, classification did not depend exclusively on depression and anxiety; however, the decline in threshold-based metrics indicates that these variables contributed meaningfully.

3.4. Sensitivity and robustness analyses

3.4.1. Analyses stratified by gender

Since men and women differ in their patterns of and susceptibility to problematic technology use (Su et al., 2020), we repeated all principal analyses separately for women (n = 3901) and men (n = 1847), re-estimating the LPA independently within each stratum (Fig. 3). BIC favored four profiles for women and men (entropy = 0.812 and 0.781, respectively). Furthermore, profile centroids correlated strongly with the full-sample solution (0.918 and 0.893, respectively). The mental-health gradient, with the Problem-Dependent profile being highest, was replicated in each group. Gender differences were minor (Cramér's V = 0.085), although the male solution was less stable (a five-profile model was a close competitor, ΔBIC = 6) and classification accuracy was higher for men (0.769) than for women (0.701), whereas macro-AUC was comparable across the strata (men 0.887; women 0.890, 95% CI [0.878, 0.903]). Therefore, the central findings were robust across the genders, with male-specific patterns warranting greater caution.

3.4.2. Adjustment for demographic composition

We further examined whether the uneven demographic composition of the profiles accounted for the mental health differences. The between-profile differences in depression, anxiety, and sleep quality remained significant (all $p < .001$), with effect sizes essentially unchanged (partial η^2 = 0.07, 0.06, 0.05) and far larger than the effect of gender (partial η^2

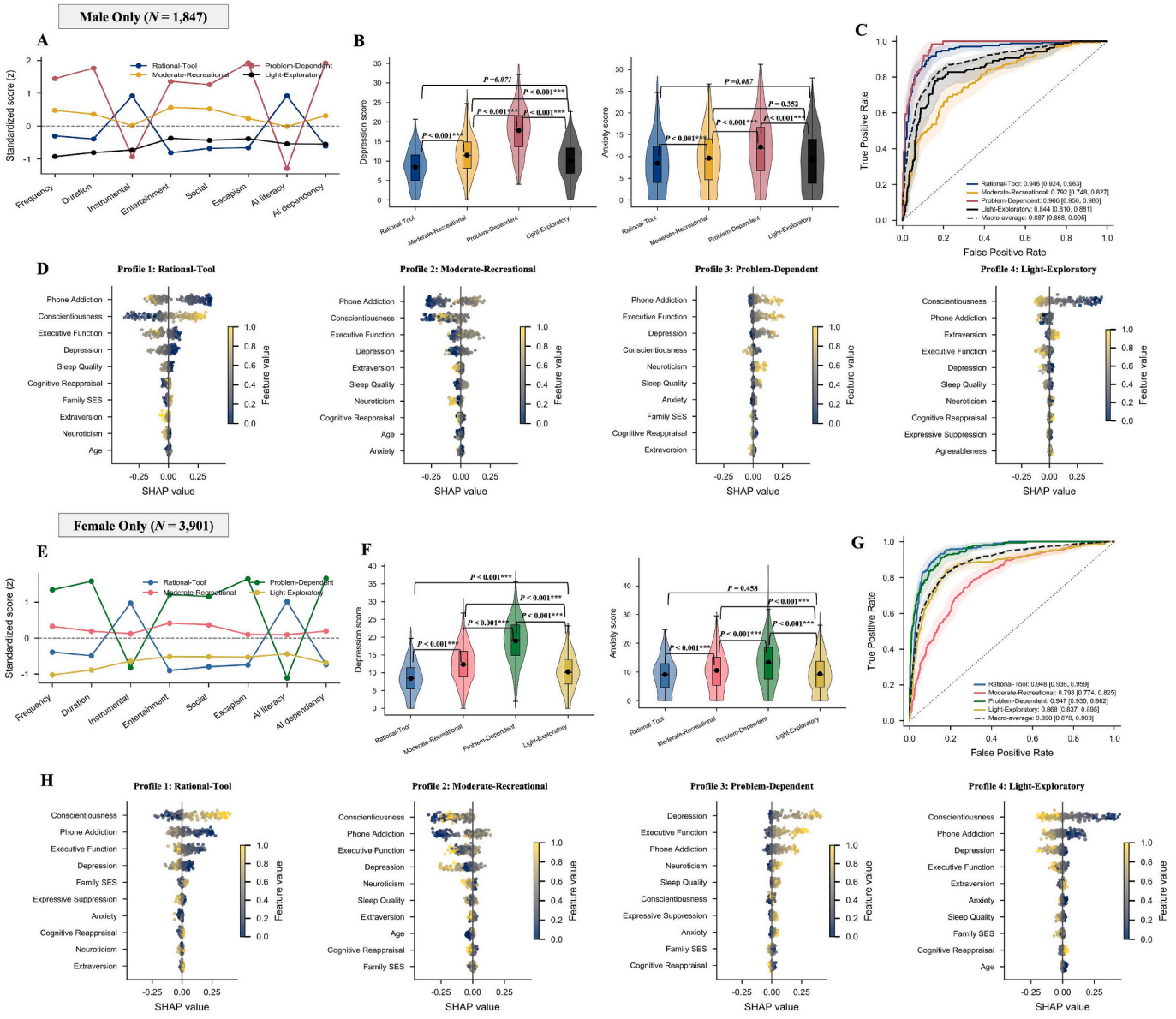


Fig. 3. Gender-stratified robustness analyses of the four-profile solution and classification.
Note. The upper half presents the male-only analyses: (A) the four-profile solution, (B) depression and anxiety distributions, (C) one-versus-rest ROC curves, and (D) class-specific SHAP beeswarm plots. The lower half presents the corresponding female-only analyses: (E) the four-profile solution, (F) depression and anxiety distributions, (G) ROC curves, and (H) SHAP beeswarm plots. Among women, the three highest-ranked SHAP features were conscientiousness, phone addiction, and executive function for Profiles 1 and 2; depression, executive function, and phone addiction for Profile 3; and conscientiousness, phone addiction, and depression for Profile 4. All profile and classification models were re-estimated separately within each gender stratum. In all SHAP beeswarm panels, point color indicates the within-feature min–max-normalized value (dark blue = low; yellow = high). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

≈ 0.01), in an ANCOVA controlling for gender, age, and family SES. The rank order and all pairwise contrasts were preserved. A multinomial logistic regression confirmed that women, younger students, and students from lower SES families were more likely to belong to the Problem-Dependent profile, yet mental health differences persisted with these covariates remaining constant (Supplementary Table S4). Finally, re-estimating the profiles from indicators residualized on gender, age, and family SES again yielded four profiles, with the Problem-Dependent profile showing the highest significance for depression and anxiety, comparable classification (accuracy = 0.774; macro-AUC = 0.875), and broadly similar leading SHAP patterns, although some rank ordering changed and conscientiousness replaced depression among the three leading features of the Moderate-Recreational profile (Supplementary Fig. S1 and Supplementary Table S5).

4. Discussion

This study identified four distinct patterns of generative AI use among 5748 Chinese college students and examined how they related to depression and anxiety. GenAI use was markedly heterogeneous, with the four profiles differing sharply in usage motivation, AI literacy, and dependency. Depression and anxiety were the highest in the Problem-Dependent profile and the lowest in the Rational-Tool profile. An exploratory random-forest analysis further showed that the profiles—defined solely by AI-use indicators—could be recovered from a separate set of psychological and demographic correlates with good discrimination (macro-average AUC = 0.894). The Problem-Dependent profile was most strongly characterized by depressive symptoms, broader technology dependence, and executive-function difficulties,

whereas the Rational-Tool profile was characterized by conscientiousness.

4.1. Discussion of results

4.1.1. Heterogeneous patterns of generative AI use

Our central finding is based on the mixed literature. Within this large sample, depression and anxiety differed systematically across the four observed profiles of GenAI use. These profiles offer a person-centered interpretation of the mixed available literature. While earlier studies reported associations of GenAI use with better and worse mental health, our findings suggest that such divergence may partly reflect differences in the mixtures of users sampled across studies. A single average estimate can obscure coexisting configurations, whereas the profile analysis distinguished lower-, intermediate-, and higher-distress patterns of GenAI use. These profiles do not present GenAI as uniformly beneficial or harmful; they show which user configurations are associated with more or less favorable mental health. These differences cannot be attributed to GenAI itself because the design is cross-sectional, but they offer an explanatory pathway for the divergent evidence and indicate configurations for longitudinal testing.

Users who used AI in an instrumental, goal-directed manner showed the lowest distress levels. Conversely, users in the Problem-Dependent profile—characterized by strong escapism motivation and lower AI literacy—showed the highest dependency and distress, consistent with the conceptualizations of excessive AI use as an emerging behavioral addiction (Yankouskaya et al., 2025). At the variable level, escapism showed the strongest associations with depression and anxiety, instrumental motivation was weakly or negatively associated with them, and AI literacy was negatively associated with both outcomes. These results do not mean that motivation or literacy independently determines profile differences; rather, they show that the profiles differed not only in use intensity but also in motivation, literacy, and dependency. Together, the findings can be interpreted through a differential-susceptibility view (Valkenburg and Peter, 2013): mental-health correlates may vary across users and configurations of use.

The four-profile structure supports the broader literature documenting heterogeneity in young people's technology use, including internet use (Wang et al., 2024), smartphone use (Yang et al., 2022), and social media use (Akbari et al., 2023). The number of profiles recovered varied across studies, reflecting differences in samples and measures. Our four-profile solution is in line with Xie et al. (2024) and Daniilidou and Antoniadou (2026), whereas Kim and Song (2022) and Wu et al. (2025) each identified three profiles. One plausible source of this divergence is that we simultaneously modeled usage motivation, AI literacy, and dependency, whereas earlier studies often relied on single-dimension measures of addiction symptoms. College students' motives for using AI may also be more varied than those of other age groups, given that this is a formative stage of identity exploration and career preparation (Arnett, 2000).

Each profile carried a distinctive configuration. The Rational-Tool profile paired high AI literacy and instrumental motivation with low dependency, a pattern that chiefly serves cognitive needs and reflects a clear goal orientation in Uses and Gratifications terms (Katz et al., 1973). The Moderate-Recreational profile (the largest group, at 33.99%) mixed instrumental, entertainment, and social motivations; because we did not directly measure need-fulfillment processes, we cannot tell from these data whether this pattern reflects compensation for unmet interpersonal needs or simply ordinary recreational engagement (Kardefelt-Winther, 2014). The Problem-Dependent profile (the smallest, at 15.00%) had the highest symptom levels and combined high frequency, high dependency, low literacy, and strong escapism motivation—a configuration that fits both the established role of escapism as a predictor of technology addiction (Chang et al., 2018; Zhou et al., 2020) and behavioral-addiction accounts of excessive AI use (Yankouskaya et al., 2025).

The Light-Exploratory profile (23.02%) deserves separate comment, because low GenAI use might seem, on its face, like the most favorable pattern. Its members had the lowest usage, motivation, and dependency of any group, yet their depression and anxiety—though below the sample average—stayed higher than the Rational-Tool profile's. Lower engagement, in other words, did not translate into the lowest symptom levels in this sample. This should not be read as evidence that using GenAI more would improve mental health; rather, it shows that frequency alone does not define these profiles. Comparable low-use subgroups in the screen-time and internet-use literature likewise tend to sit in a middle position rather than a clearly protective one (Moreno et al., 2022; Wang et al., 2024).

4.1.2. Associations between use patterns and depression and anxiety

Depression and anxiety peaked in the Problem-Dependent profile and were the lowest in the Rational-Tool profile, which is consistent with studies on adolescent internet addiction (Wang et al., 2024) and smartphone addiction (Kim et al., 2023). The profile differences were small-to-moderate (partial $\eta^2 = 0.07$ for depression and 0.06 for anxiety; Cohen, 1992). However, magnitude rather than statistical significance should guide data interpretation with a sample this large (Sullivan and Feinn, 2012). The mean BDI-II score differed by 8.7 points (16.87 vs. 8.14), spanning the mild and minimal symptom ranges. Small profile-level associations may matter when they accumulate or operate at a scale (Abelson, 1985; Funder and Ozer, 2019; Götz et al., 2022; Rose, 1985), but these cross-sectional differences should not be interpreted as individual-level clinical effects or causal evidence.

Several mechanisms could underlie this gradient, although none can be confirmed with cross-sectional data. The social displacement hypothesis (Kraut et al., 1998) suggests that time spent in digital interaction may crowd out face-to-face contact, with the attendant loss of social support being associated with more depressive symptoms. Although the hypothesis is not universal (Beyens et al., 2020), displacement may be more pronounced when use is motivated by escapism (Meier and Reinecke, 2021). Problem-Dependent users may immerse themselves in AI interaction to sidestep real-world stress. Such use may provide temporary relief without resolving the underlying problems or building adaptive coping skills. Problematic AI use may similarly function as a maladaptive coping pattern that co-occurs with anxiety symptoms (Stavropoulos et al., 2017). Additionally, highly dependent users may experience withdrawal-like anxiety when access is blocked (Yankouskaya et al., 2025). Longitudinal research on earlier technologies documented bidirectional relationships—distress predicts dependency, which in turn exacerbates distress (Coyne et al., 2020; Houghton et al., 2018)—and similar reciprocal patterns may plausibly characterize AI dependency, although direct longitudinal evidence for GenAI does not yet exist.

These dynamics may be amplified by features that set GenAI apart from earlier technologies. Unlike conventional social media, conversational agents offer highly personalized, immediately responsive interaction that can function as an always-available "conversation partner"—something that may hold particular appeal for students who are lonely or distressed. Yankouskaya et al. (2025) described problematic ChatGPT use in terms of instant gratification, parasocial bonding, loss of control, and withdrawal-like symptoms, features that parallel established behavioral addictions. The same affordances that make GenAI an accessible source of support for some users may make it an especially potent focus of dependence for others—an ambivalence consistent with the divergent patterns we observed across profiles.

Escapism showed the strongest associations with depression and anxiety, consistent with evidence that escapism-motivated social media use predicts mental-health problems more strongly than instrumental use does (Bányai et al., 2017). Escape theory (Baumeister, 1990) offers one framework for this pattern: escape-motivated activities narrow attention and reduce aversive self-awareness. This distinguishes escapism from ordinary recreational use—what matters is not

enjoyment or high frequency per se, but using GenAI to disengage from unwanted thoughts or emotions. Whether this motive precedes distress or develops in response to it is a question for future longitudinal work.

The inverse association between AI literacy and distress also merits attention, and several mechanisms could plausibly explain it. Users with higher literacy may be better able to critically evaluate AI-generated content, recognize its limitations, and calibrate how much they rely on it—reducing the uncritical, immersive engagement typical of problematic use (Long and Magerko, 2020; Ng et al., 2021). Literacy is also linked to greater self-efficacy and a more goal-directed orientation (Laupichler et al., 2022), whereas low literacy tends to carry a greater psychological burden during technology use (Bansal and Choudhary, 2024). Within the UTAUT framework, technology adoption depends on users' expectations and facilitating conditions (Venkatesh et al., 2003), so AI literacy may shape how effectively people engage with GenAI overall. Because literacy is likely more modifiable than stable traits such as neuroticism, its prospective role deserves evaluation in longitudinal and intervention research.

4.1.3. Psychological correlates of profile membership

The executive-function difficulties concentrated in the Problem-Dependent profile deserve attention, since executive function underlies self-regulation more broadly (Hofmann et al., 2012). Deficits here could make it harder to control use, switch flexibly between activities, and inhibit impulses toward dependence. Neurobiological work offers one plausible mechanism: internet gaming disorder is linked to reduced prefrontal function and weaker cognitive-control connectivity (Dong and Potenza, 2014), and the I-PACE model similarly holds that executive deficits impair the monitoring and regulation of technology use (Brand et al., 2019). This study cannot determine whether such deficits precede or follow problematic AI use.

The random forest and SHAP results add a multivariate perspective: the profiles were classified with good overall discrimination (macro-average AUC = 0.894), and depression contributed strongly to identifying the Problem-Dependent profile. Because depression and anxiety were also outcomes in the between-profile comparisons, their prominence in SHAP is convergent rather than independent predictive evidence. The cross-sectional design cannot tell us the direction of this relationship: depressed individuals may turn to AI for readily available comfort, problematic use may accompany higher symptoms, or the two may reinforce each other over time (Kardefelt-Winther, 2014). Smartphone addiction was another leading contributor, which suggests the Problem-Dependent profile may reflect broader problematic technology use rather than something specific to AI. This fits the known co-occurrence of problematic technology-use behaviors (Hussain et al., 2020; Toklu Baloglu and Caferoglu Akin, 2024) and shared mechanisms involving reward processing, impulse control, and habit formation (Brand et al., 2019; Liu et al., 2018).

Associations of neuroticism and conscientiousness with profile membership were consistent with the Person-Environment Fit Theory (Edwards, 1991) and prior research on personality and problematic technology use (Marciano et al., 2022). Neuroticism characterized the Problem-Dependent profile, whereas conscientiousness characterized the Rational-Tool profile. Emotion regulation showed a parallel pattern: Problem-Dependent users reported less cognitive reappraisal and more expressive suppression (Gross and John, 2003), which is consistent with evidence showing that reappraisal is negatively related to internet addiction and that the association between suppression and internet addiction may operate through negative affect (Liang et al., 2021). These findings support further longitudinal research on whether emotion-regulation difficulties precede or follow problematic GenAI use.

4.2. Significance

Our findings illustrate the added value of a person-centered approach theoretically and methodologically. Whereas many variable-centered analyses summarize average associations between variables, LPA identified recurring configurations of usage behavior, motivation, AI literacy, and dependency. The analysis showed how profiles with different distress levels can coexist within one population by shifting the unit of inference from individual variables to multidimensional configurations. Therefore, the findings are consistent with user heterogeneity as one plausible contributor to divergent aggregate results, rather than an empirical explanation of why prior studies differ.

Practically, rather than supporting a uniform message to decrease AI use, the profiles generate hypotheses about potentially different support needs. For the highest-distress Problem-Dependent profile, future interventions could test combinations of adaptive emotion-regulation training (favoring cognitive reappraisal over expressive suppression; Gross and John, 2003), AI-literacy education, and executive-function support. Additionally, elevated depression and anxiety should be addressed through established screening and referral pathways. For the Moderate-Recreational profile, future work could examine whether strengthening real-world social support helps AI supplement interpersonal connection rather than substitute for it (Kardefelt-Winther, 2014). Guidance for Light-Exploratory users might emphasize informed, purposeful use, whereas support for Rational-Tool users might focus on sustaining goal-directed engagement. These profile-specific hypotheses complement prevention at the populational level (Rose, 1985); however, whether profile-matched interventions outperform blanket restrictions remains to be tested.

The findings may also inform educational policy. Because higher AI literacy was associated with lower distress and was more characteristic of the lower-distress profiles, universities have a rationale for providing AI-literacy education that extends beyond technical proficiency to the critical evaluation of AI-generated content, awareness of its limitations, ethical reflection, and healthy usage boundaries (Naamati-Schneider and Alt, 2025). Given the co-occurrence of problematic AI use and smartphone addiction, such education may fit best within a broader digital-health framework. Moreover, future research should evaluate whether multidimensional indicators—motivation, literacy, and dependency rather than frequency alone—improve the identification of students who may benefit from further assessment; still, profile membership should not substitute for validated clinical screening.

Finally, pairing a person-centered model of multidimensional indicators with interpretable machine learning offers an approach that could transfer to other work characterizing GenAI use profiles and their external correlates. This exploratory, theory-guided strategy may suit emerging technologies particularly well, where explanatory theory is still catching up. Future research should test it longitudinally and examine whether it replicates across technologies, cultures, and developmental stages.

4.3. Limitations and future directions

This study has several limitations. First, its cross-sectional design precludes causal inference and leaves unresolved the temporal direction between profile membership and mental health. Longitudinal designs—latent transition analysis in particular—are needed to trace how profiles form and change over time, and how usage patterns and symptoms relate to each other as they do.

Second, excluding students receiving psychotherapy or psychiatric medication likely removed some participants with more severe symptoms and thus might have truncated the upper tail of the symptom distributions. Such exclusions can make study samples less representative of real-world depression and anxiety populations (Blanco et al., 2008; de Graaf et al., 2000; Galbaud du Fort et al., 1993; Hoertel et al., 2012; Wisniewski et al., 2009; Zimmerman et al., 2002). However, the direction and magnitude of the resulting selection bias are uncertain because excluded students may differ in symptom severity and GenAI-use patterns. Hence, replication in samples that include clinically

treated individuals is warranted. Furthermore, GenAI use among young people managing diagnosed conditions deserves direct evaluation.

Third, the sample was drawn from four universities in western China, predominantly represented by females (67.86%) and reflecting recruitment largely from teacher-training institutions. Female-skewed samples are common in the literature (Akbari et al., 2023; Aslan and Koç, 2026; Cui et al., 2023), and our covariate-adjusted and gender-stratified analyses indicated that the central findings were robust across the genders, with the four-profile structure and the mental-health gradient replicating in both strata. Nonetheless, some gender differences emerged, and the male solution was less stable; thus, male-specific patterns warrant caution. Because men and women also differ in their susceptibility to problematic technology use and in the forms it takes (Su et al., 2019, 2020), future research should employ more gender-balanced and geographically diverse samples.

Fourth, two measurement limitations apply. All constructs were assessed in a self-reporting manner, with its attendant social-desirability and recall biases. Therefore, objective behavioral data such as usage logs, together with neurocognitive testing and qualitative methods, would strengthen measurement and deepen understanding of the experiences underlying each pattern. Additionally, although the AI-dependency scale was adapted from validated measures and performed well psychometrically in this sample, it was grounded in the components model of behavioral addiction and may not fully capture GenAI-specific features such as anthropomorphism and perceived companionship. Thus, the development and validation of GenAI-specific dependency measures are a priority for future work.

Fifth, the study did not differentiate among GenAI tool types, which vary substantially in their psychological affordances—from text-based agents serving informational and emotional-support functions to companion applications designed to elicit anthropomorphism and parasocial attachment (Epley et al., 2007; Skjuve et al., 2021). Companion-oriented use may be especially consequential, as emotional dependence on social chatbots such as Replika is linked to harm, similar to that which can arise in human relationships (Pentina et al., 2023; De Freitas et al., 2024; Laestadius et al., 2024). Aggregating across such tools may obscure tool-specific associations, paralleling critiques of undifferentiated "screen time" (Kaye et al., 2020). Therefore, the present profiles should be read as an aggregate description of use patterns rather than a tool-specific structure. A recent meta-analysis has found larger mental health effects for socially oriented than for task-oriented chatbots (Zhang et al., 2025). Since interaction type may matter in its own right, future research should measure tool and interaction type explicitly and test whether these profiles replicate or split into more tool-specific ones.

5. Conclusions

This study identified four generative AI use profiles among Chinese college students (Rational-Tool, Moderate-Recreational, Problem-Dependent, and Light-Exploratory), which showed a clear gradient in mental health, with depression and anxiety showing the highest and lowest levels in the Problem-Dependent and Rational-Tool profiles, respectively. An exploratory machine-learning analysis converged on the same pattern: from independent psychological and demographic correlates, the Problem-Dependent profile was set apart chiefly by symptom burden, wider technology dependence, and weaker executive control, whereas the Rational-Tool profile was set apart by conscientiousness and AI literacy. Taken together, the mental-health correlates of GenAI use depend on the configuration of use—motivation, literacy, and dependency—rather than on frequency alone. They also offer a person-centered explanation for why studies that average across heterogeneous users may report mixed associations between GenAI use and mental health. The current study provides an empirical basis for more targeted prevention by identifying higher-risk configurations rather than treating all use as equivalent.

CRediT authorship contribution statement

Haoyang Shi: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Tongyi Zhang:** Writing – original draft, Methodology, Formal analysis, Data curation. **Zihao Wang:** Methodology, Investigation, Data curation. **Xiaolong Yang:** Writing – review & editing, Funding acquisition. **Zixuan Shao:** Writing – review & editing. **Xiaofeng Ma:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Funding statement

This work was supported by the Gansu Provincial Disease Prevention and Control Research Program (No. GSJKKY2025-19 [to Xiaolong Yang]) and the Gansu Provincial Health Industry Research Project (No. GSWSQN2025-25 [to Xiaolong Yang]).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank all participants for taking part in this study.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jad.2026.122428.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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